

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a logically isolated network inside AWS where you can launch your resources (EC2, RDS, Load Balancers, etc.).

It gives you:

- Full control over IP addressing
- Subnet creation
- Route tables
- Internet access configuration
- Security boundaries

👉 Imagine AWS is a huge building, VPC is your private floor inside that building. Only you control what enters and leaves.

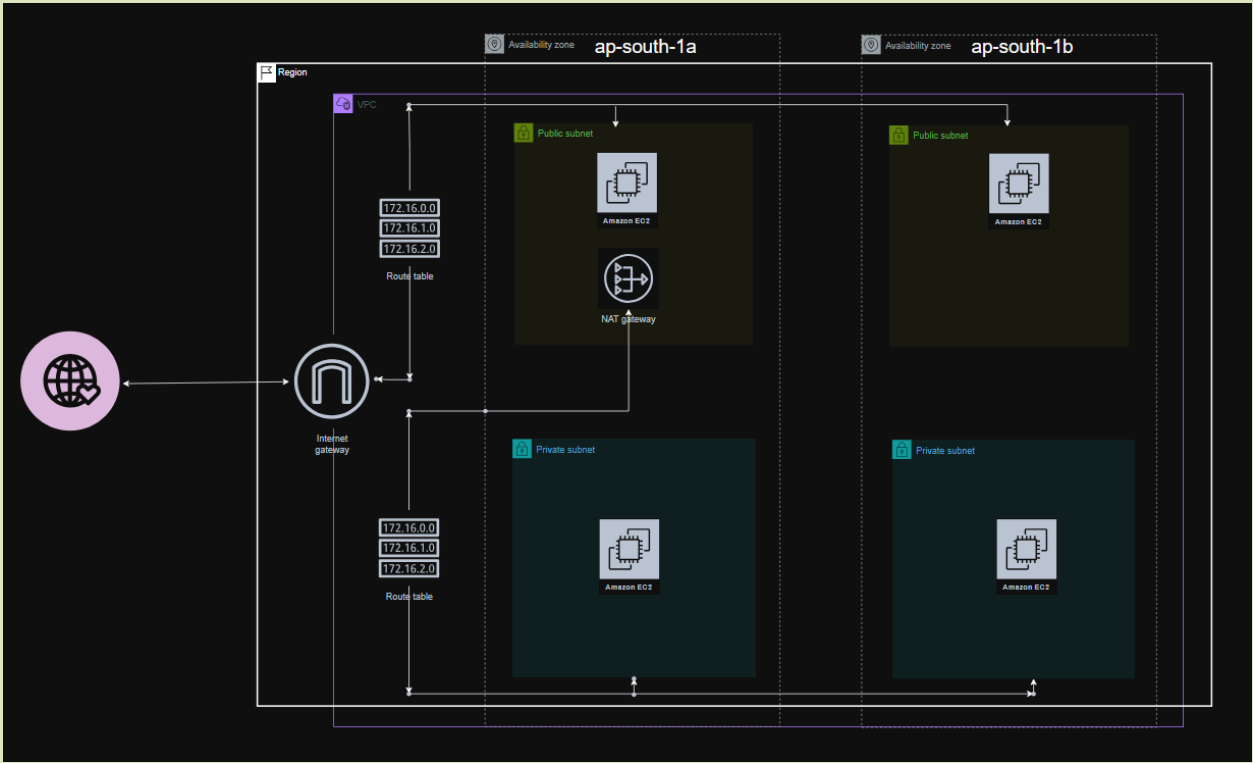
Project Overview

In this project, we design a **production-ready AWS network architecture** with:

- 1 Custom VPC
- 2 Public Subnets
- 2 Private Subnets
- Internet Gateway
- NAT Gateway
- Route Tables
- 4 EC2 Instances (1 in each subnet)

This is a standard architecture used in real-world applications.

Architecture Diagram



SECURE MULTI-TIER VPC ARCHITECTURE

Why Do We Need This Architecture?

Without proper network design:

- Servers become publicly exposed
- Security risks increase
- No separation between frontend and backend
- Hard to scale

With this architecture:

- Public servers are accessible
- Private servers are protected
- Internet access is controlled
- Architecture is scalable

How This Architecture Works (Simple Flow)

Public Subnet Flow

User → Internet → Internet Gateway → Public EC2

Public instances have Public IP.

Private Subnet Flow

Private EC2 → Private Route Table → NAT Gateway → Internet Gateway → Internet

Private instances:

- Do NOT have public IP
- Cannot be accessed directly from internet
- Can access internet only for updates/downloads

→ Let's do practice

Step 1 – Create VPC

Go to:

AWS Console → VPC → Create VPC

Configuration:

- Name: My-VPC
- IPv4 CIDR: 10.0.0.0/16
- Tenancy: Default

Click **Create VPC**

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

My-VPC

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

Step 2 – Create Subnets

Create 4 Subnets inside the VPC.

Public Subnet 1 → 10.0.1.0/24

Public Subnet 2 → 10.0.2.0/24

Private Subnet 1 → 10.0.3.0/24

Private Subnet 2 → 10.0.4.0/24

Enable **Auto Assign Public IP** only for public subnets.

Go to:

VPC → Subnets → Create Subnet

Select your VPC.

Create:

VPC

VPC ID

Create subnets in this VPC.

vpc-0b9e2f99d3e7ba21e (My-VPC)

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Public Subnet 1

- Name: Public-Subnet-1
- AZ: ap-south-1a
- CIDR: 10.0.1.0/24

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public-Subnet-1

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az1 (ap-south-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

256 IPs

< > ^ v

Public Subnet 2

- Name: Public-Subnet-2
- AZ: ap-south-1b
- CIDR: 10.0.2.0/24

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs

< > ^ v

Private Subnet 1

- Name: Private-Subnet-1
- AZ: ap-south-1a
- CIDR: 10.0.3.0/24

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs

< > ^ v

Private Subnet 2

- Name: Private-Subnet-2
- AZ: ap-south-1b
- CIDR: 10.0.4.0/24

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs

< > ^ v

Enable Auto-Assign Public IP (IMPORTANT)

Go to:

VPC → Subnets → Select Public-Subnet-1 → Edit Subnet Settings → Enable:

☒ Auto-assign public IPv4 address

Do same for → Public-Subnet-2

Do NOT enable this for private subnets.

Subnets (1/7) [Info](#)

☐	Name	Subnet ID	State	VPC
☐	-	[REDACTED]	Available	vpc-06b12fa9f613
☐	-	[REDACTED]	Available	vpc-06b12fa9f613
☐	-	[REDACTED]	Available	vpc-06b12fa9f613
☒	Public-Subnet-1	subnet-0de3fc744e9b04d4d	Available	vpc-0b9e2f99d3e
☐	Public-Subnet-2	subnet-0f9a3cb391b26ab5e	Available	vpc-0b9e2f99d3e
☐	Private-Subnet-1	subnet-040e15a81cfa6e7f2	Available	vpc-0b9e2f99d3e
☐	Private-Subnet-2	subnet-0fd727ed6b7f29c86	Available	vpc-0b9e2f99d3e

Last updated 1 minute ago

Actions

View details

Create flow log

Edit subnet settings

Edit IPv6 CIDRs

Edit network ACL association

Edit route table association

Edit CIDR reservations

Share subnet

Manage tags

Delete subnet

Edit subnet settings [Info](#)

Subnet

Subnet ID	Name
subnet-0de3fc744e9b04d4d	Public-Subnet-1

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)
Option disabled because no customer owned pools found.

Step 3 – Create Internet Gateway

Go to:

VPC → Internet Gateways → Create

- Name: My-IGW

After creation:

Click → Attach to VPC → Select **My-VPC**

Now public subnets can connect to internet (after route table config).

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	My-IGW	Remove

You can add 49 more tags.

igw-0d48008927579788c / My-IGW

Details [info](#)

Internet gateway ID

igw-0d48008927579788c

State

Detached

VPC ID

-

Owner

968626770773

Actions

Attach to VPC

Detach from VPC

Manage tags

Delete

Tags (1)

Search tags

Key	Value
Name	My-IGW

Manage tags

Attach to VPC (igw-0d48008927579788c) [info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

Q vpc-0b9e2f99d3e7ba21e

Use: "vpc-0b9e2f99d3e7ba21e"

vpc-0b9e2f99d3e7ba21e - My-VPC

Cancel

Attach internet gateway

Step 4 – Configure Public Route Table

Go to:

VPC → Route Tables → Create

- Name: Public-Route-Table
- VPC: My-VPC

Create route table [info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

Public-Route-Table

VPC

The VPC to use for this route table.

vpc-0b9e2f99d3e7ba21e (My-VPC)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q Public-Route-Table

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Now:

Edit Routes → Add Route

Destination: 0.0.0.0/0

Target: Select your Internet Gateway

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Associate Public Subnets

Go to:

Subnet Associations → Edit

Select:

- Public-Subnet-1
- Public-Subnet-2

rtb-0c47cbc721dc79f6e / Public-Route-Table

Details

Route table ID: rtb-0c47cbc721dc79f6e

VPC: vpc-0b9e2f99d3e7ba21e | My-VPC

Main: No

Owner ID: 968626770773

Explicit subnet associations: -

Edge associations: -

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0d48008927579788c	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (2/4)

Q Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒ Public-Subnet-1	subnet-Ode3fc744e9b04d4d	10.0.1.0/24	-	Main (rtb-024b55228fe562c5e)
☒ Public-Subnet-2	subnet-Of9a3cb391b26ab5e	10.0.2.0/24	-	Main (rtb-024b55228fe562c5e)
☐ Private-Subnet-1	subnet-O40e15a81cfa6e7f2	10.0.3.0/24	-	Main (rtb-024b55228fe562c5e)
☐ Private-Subnet-2	subnet-Ofd727ed6b7f29c86	10.0.4.0/24	-	Main (rtb-024b55228fe562c5e)

Selected subnets

subnet-Ode3fc744e9b04d4d / Public-Subnet-1 ✕ subnet-Of9a3cb391b26ab5e / Public-Subnet-2 ✕

[Cancel](#) [Save associations](#)

Now public instances can access internet.

Step 5 – Create NAT Gateway (For Private Subnets)

⚠ **NAT** must be created in **PUBLIC** subnet.

Go to:

VPC → NAT Gateways → Create

Configuration:

- Name: My-NAT
- Subnet: Public-Subnet-1
- Connectivity: Public
- Allocate Elastic IP (very important)

✓ Elastic IP address 13.206.34.34 (eipalloc-04ea44afd04106904) allocated.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

My-NAT

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.

subnet-Ode3fc744e9b04d4d (Public-Subnet-1)

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public

☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

eipalloc-04ea44afd04106904 [Allocate Elastic IP](#)

Wait until status becomes Available.

Step 6 – Configure Private Route Table

Go to:

VPC → Route Tables → Create

- Name: Private-Route-Table
- VPC: My-VPC

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Private-Route-Table

VPC
The VPC to use for this route table.

vpc-0b9e2f99d3e7ba21e (My-VPC)

Now:

Edit Routes → Add Route

Destination: 0.0.0.0/0

Target: Select NAT Gateway

Edit routes

Destination	Target	Status
10.0.0.0/16	local	✓ Active
0.0.0.0/0	NAT Gateway	-
	nat-011cd7a808f31b268	
	Use: "nat-011cd7a808f31b268"	
	nat-011cd7a808f31b268 (My-NAT)	

Add route

Now private instances can access internet securely.

Associate Private Subnets

Go to:

Subnet Associations → Edit

Select:

- Private-Subnet-1
- Private-Subnet-2

☰ [VPC](#) > [Route tables](#) > [rtb-0d5801332bb447ce0](#) > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

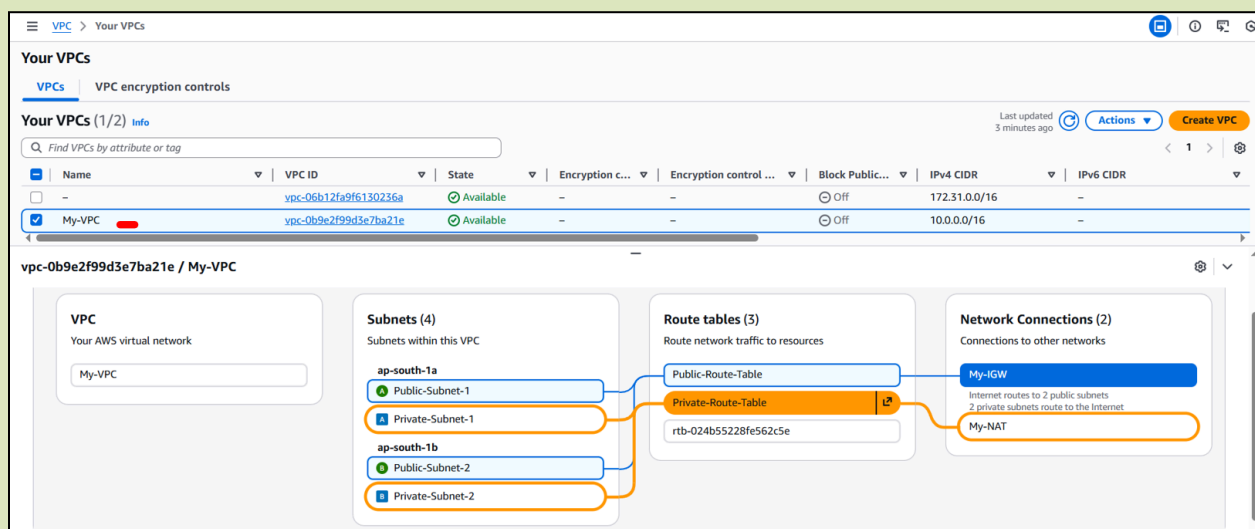
Available subnets (2/4)

🔍 *Filter subnet associations*

☐	Name	Subnet ID	IPv4 CIDR
☐	Public-Subnet-1	subnet-0de3fc744e9b04d4d	10.0.1.0/24
☐	Public-Subnet-2	subnet-0f9a3cb391b26ab5e	10.0.2.0/24
☒	Private-Subnet-1	subnet-040e15a81cfa6e7f2	10.0.3.0/24
☒	Private-Subnet-2	subnet-0fd727ed6b7f29c86	10.0.4.0/24

Selected subnets

[subnet-040e15a81cfa6e7f2 / Private-Subnet-1](#) ✕ [subnet-0fd727ed6b7f29c86 / Private-Subnet-2](#) ✕



This is resource map of your vpc.

Step 6: Create Security Groups

You need at least 2 SGs:

Public-Instance-SG

Allow:

- SSH (22) → My IP or 0.0.0.0/0
- HTTP (80) → 0.0.0.0/0
- ALL ICMP-IPV4 → 0.0.0.0/0

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

Public-Instance-SG

Name cannot be edited after creation.

Description Info

Allows SSH access to developers

VPC Info

vpc-0b9e2f99d3e7ba21e (My-VPC)

Inbound rules Info

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
SSH	TCP	22	Anywh... Q 0.0.0.0/0 0.0.0.0/0 X	Delete
HTTP	TCP	80	Anywh... Q 0.0.0.0/0 0.0.0.0/0 X	Delete
All ICMP - IPv4	ICMP	All	Anywh... Q 0.0.0.0/0 0.0.0.0/0 X	Delete

Private-Instance-SG

Allow:

- SSH (22) → My IP or 0.0.0.0/0
- HTTP (80) → 0.0.0.0/0
- ALL ICMP-IPV4 → 0.0.0.0/0

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Private-Instance-SG

Name cannot be edited after creation.

Description [Info](#)

Allows SSH access to developers

VPC [Info](#)

vpc-0b9e2f99d3e7ba21e (My-VPC)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
SSH	TCP	22	Anywhere...	0.0.0.0/0	Delete
HTTP	TCP	80	Anywhere...	0.0.0.0/0	Delete
All ICMP - IPv4	ICMP	All	Anywhere...	0.0.0.0/0	Delete

Step 7: Launch EC2 Instances

2 Public Instances

- Select Public Subnet
- Auto-assign Public IP → Enabled
- Attach Public SG

Network settings [Info](#)

VPC - required [Info](#)

vpc-0b9e2f99d3e7ba21e (My-VPC)

Subnet [Info](#)

subnet-0de3fc744e9b04d4d Public-Subnet-1

VPC: vpc-0b9e2f99d3e7ba21e Owner: 968626770773 Availability Zone: ap-south-1a (aps1-az1)

Zone type: Availability Zone IP addresses available: 250 CIDR: 10.0.1.0/24

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

Public-Instance-SG sg-0ce2b40f4c4ae38cd

VPC: vpc-0b9e2f99d3e7ba21e

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Advanced network configuration

▼ Network settings

Info

VPC - required

Info

vpc-0b9e2f99d3e7ba21e (My-VPC)

10.0.0.0/16

Subnet

Info

subnet-0f9a3cb391b26ab5e

Public-Subnet-2

VPC: vpc-0b9e2f99d3e7ba21e

Owner: 968626770773

Availability Zone: ap-south-1b (aps1-az3)

Zone type: Availability Zone

IP addresses available: 251

CIDR: 10.0.2.0/24

Create new subnet

Auto-assign public IP

Info

Enable

Additional charges apply

when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups

Info

Select security groups

Public-Instance-SG

sg-0ce2b40f4c4ae38cd

VPC: vpc-0b9e2f99d3e7ba21e

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

2 Private Instances

- Select Private Subnet
- Auto-assign Public IP → Disabled
- Attach Private SG

▼ Network settings

Info

VPC - required

Info

vpc-0b9e2f99d3e7ba21e (My-VPC)

10.0.0.0/16

Subnet

Info

subnet-040e15a81cfa6e7f2

Private-Subnet-1

VPC: vpc-0b9e2f99d3e7ba21e

Owner: 968626770773

Availability Zone: ap-south-1a (aps1-az1)

Zone type: Availability Zone

IP addresses available: 251

CIDR: 10.0.3.0/24

Create new subnet

Auto-assign public IP

Info

Disable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups

Info

Select security groups

Private-Instance-SG

sg-0db25a209209e46f7

VPC: vpc-0b9e2f99d3e7ba21e

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0b9e2f99d3e7ba21e (My-VPC) 10.0.0.0/16

Subnet [Info](#)

subnet-0fd727ed6b7f29c86 Private-Subnet-2
 VPC: vpc-0b9e2f99d3e7ba21e Owner: 968626770773 Availability Zone: ap-south-1b (aps1-az3)
 Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.4.0/24

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

Private-Instance-SG sg-0db25a209209e46f7
 VPC: vpc-0b9e2f99d3e7ba21e

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► **Advanced network configuration**

Step 8: Test Connectivity

Test Public Instance

SSH using public IP:

ssh -i key.pem ec2-user@<public-ip>

```
ubuntu@ip-10-0-1-124: ~
user@Lalit-Thakur MINGW64 ~/Desktop
$ ssh -i "Dev-Key.pem" ubuntu@13.126.103.205
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Fri Feb 20 09:01:00 UTC 2026

System load:  0.0           Processes:            107
Usage of /:   25.9% of 6.71GB Users logged in:          0
Memory usage: 20%          IPv4 address for enx0: 10.0.1.124
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
```

You are connected to your public instance because internet is publicly available to you public instances due to internet gateway

Run:
ping google.com

```
ubuntu@ip-10-0-1-124: ~  
ubuntu@ip-10-0-1-124:~$ ping google.com  
PING google.com (192.178.211.138) 56(84) bytes of data.  
64 bytes from 1cbomp-in-f138.1e100.net (192.178.211.138): icmp_seq=1 ttl=107 time=2.46 ms  
64 bytes from 1cbomp-in-f138.1e100.net (192.178.211.138): icmp_seq=2 ttl=107 time=2.44 ms  
64 bytes from 1cbomp-in-f138.1e100.net (192.178.211.138): icmp_seq=3 ttl=107 time=2.45 ms  
64 bytes from 1cbomp-in-f138.1e100.net (192.178.211.138): icmp_seq=4 ttl=107 time=2.49 ms  
64 bytes from 1cbomp-in-f138.1e100.net (192.178.211.138): icmp_seq=5 ttl=107 time=2.45 ms  
^C  
--- google.com ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 2.440/2.458/2.489/0.017 ms  
ubuntu@ip-10-0-1-124:~$ |
```

Test Private Instance

SSH into public instance first (bastion method).

Then from public instance:

ssh ec2-user@<private-ip>

```
ubuntu@ip-10-0-1-124: ~  
ubuntu@ip-10-0-1-124:~$ vi key.pem  
ubuntu@ip-10-0-1-124:~$  
ubuntu@ip-10-0-1-124:~$ ls -l  
total 4  
-rw-rw-r-- 1 ubuntu ubuntu 1675 Feb 20 09:05 key.pem  
ubuntu@ip-10-0-1-124:~$ chmod 400 key.pem  
ubuntu@ip-10-0-1-124:~$  
ubuntu@ip-10-0-1-124:~$ ls -l  
total 4  
-r----- 1 ubuntu ubuntu 1675 Feb 20 09:05 key.pem  
ubuntu@ip-10-0-1-124:~$ |
```

To SSH into your private instances, you must use a **public instance configured as a Bastion Host**.

You cannot directly SSH into private instances because they do **not have a public IP address assigned**. They are provisioned with only **private IPv4 addresses**, which means they are accessible **only within the VPC's private network**.

Since all instances are deployed within the same **VPC (Virtual Private Cloud)** in AWS, they can communicate with each other using their private IP addresses through the VPC's internal networking.

Therefore, the correct approach is:

1. SSH from your local machine to the **public EC2 instance (Bastion Host)** using its public IP.
2. From the Bastion Host, SSH into the **private EC2 instances** using their private IP addresses.

This works because:

- The Bastion Host resides in a **public subnet** with an Internet Gateway.
- The private instances reside in a **private subnet** without public IPs.
- Internal communication within the same VPC is allowed by default (subject to Security Group and NACL rules).

Instance summary for i-0f755560d102ad150 (Private-Instance-1) [Info](#)

Updated less than a minute ago

Instance ID

 i-0f755560d102ad150

IPv6 address

–

Hostname type

IP name: ip-10-0-3-132.ap-south-1.compute.internal

Answer private resource DNS name

–

Auto-assigned IP address

–

IAM role

–

Public IPv4 address

–

Instance state

 Running

Private IP DNS name (IPv4 only)

 ip-10-0-3-132.ap-south-1.compute.internal

Instance type

t2.micro

VPC ID

 vpc-0b9e2f99d3e7ba21e (My-VPC) [↗](#)

Subnet ID

 subnet-040e15a81cfa6e7f2 (Private-Subnet-1) [↗](#)

Private IPv4 addresses

 10.0.3.132 

Public DNS

–

Elastic IP addresses

–

AWS Compute Optimizer finding

 Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#) [↗](#)

Auto Scaling Group name

–

```
ubuntu@ip-10-0-3-132: ~  
ubuntu@ip-10-0-1-124:~$ ssh -i key.pem ubuntu@10.0.3.132  
The authenticity of host '10.0.3.132 (10.0.3.132)' can't be established.  
ED25519 key fingerprint is SHA256:h3qtC0ufPS9PQjRGq2q80/KsCg8//hyf4xBoIx1tVNY.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.3.132' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)
```

Now test:

ping google.com

It should work through NAT Gateway.

```
ubuntu@ip-10-0-3-132:~$ ping google.com  
PING google.com (192.178.211.113) 56(84) bytes of data.  
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=1 ttl=106 time=3.31 ms  
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=2 ttl=106 time=3.07 ms  
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=3 ttl=106 time=3.06 ms  
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=4 ttl=106 time=3.05 ms  
^C  
--- google.com ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3004ms  
rtt min/avg/max/mdev = 3.047/3.123/3.312/0.109 ms  
ubuntu@ip-10-0-3-132:~$
```

As you can see, the instance is able to ping **google.com**, which confirms that internet access is available.

However, this internet connectivity is **not through the Internet Gateway directly**.

The instance is deployed inside a **private subnet**, which does not have a route to the Internet Gateway. Instead, its route table points to a **NAT Gateway** located in a public subnet.

The traffic flow works as follows:

1. The private instance sends outbound traffic (e.g., ping to Google).
2. The route table of the private subnet forwards this traffic to the **NAT Gateway**.

3. The NAT Gateway, which has a public IP and is attached to an **Internet Gateway**, forwards the request to the internet.
4. The response returns to the NAT Gateway.
5. The NAT Gateway sends the response back to the private instance.

This confirms:

-  The private instance has outbound internet access.
-  The private instance does not have direct inbound internet access.
-  Internet connectivity is provided via the **NAT Gateway**, not directly via the Internet Gateway.

This is the correct and secure architecture for private subnet design in AWS.

What Happens Internally?

Public instance flow:

Instance → Route Table → IGW → Internet

Private instance flow:

Instance → Private Route Table → NAT → IGW → Internet

Security Design

- Private instances have no public IP
- Internet access only via NAT
- Security groups restrict inbound traffic
- Multi-AZ setup improves availability

This design follows AWS best practices.